SS-C-208c
NOVEMBER 21, 1960
SUPERSEDING
Fed. Spec. SS-C-208b
April 9, 1954

FEDERAL SPECIFICATION

CEMENT, PORTLAND-POZZOLAN

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. CLASSIFICATION

1.1 Types.—The portland-pozzolan cement covered by this specification shall be of the following types, as specified:

Type I.—Portland-pozzolan cement non-air-entraining

Type IA.—Portland-pozzolan cement, air-entraining

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitations for bids, form a part of this specification:

Federal Specifications:

SS–C–192—Cement, Portland.
UU–S–48—Sacks, Shipping, Paper.

Federal Standards:

Fed. Test Method Std. No. 158 — Cements, Hydraulic, Sampling, Inspection and Testing.
Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Portland cement or portland-cement clinker used in the portland-pozzolan cement shall conform to the requirements for chemical composition for type I, Federal Specification SS–C–192. Pozzolans used in portland-pozzolan cement shall conform to the requirements specified in 3.5.

3.2 The percentage of pozzolan in the portland-pozzolan cement shall be not less than

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15 percent nor more than 35 percent, by weight. The manufacturer shall state the source, amount, and composition of the pozzolan used in the finished portland-pozzolan cement. The amount of pozzolan in the finished cement shall not vary more than ± 5.0 percentage points by weight from the amount stated by the manufacturer.

TABLE I.—*Chemical requirements*

Test	Type I	Type IA
	Percent	*Percent*
Sulfur trioxide (SO_3), maximum	2.5	2.5
Moisture content, maximum	3.0	3.5
Ignition loss, maximum	3.0	3.0

3.4 Air-entraining addition. — When air-entraining cement is specified, there shall be interground with the clinker and addition of such a nature and in such an amount that the resulting product complies with the applicable requirements of this specification. At the request of the purchaser, the manufacturer shall furnish a written statement giving the source, amount and brand name of the addition used. At the request of the purchaser, the manufacturer shall also furnish test data (note) acceptable to the purchaser showing that air-entraining portland cement that contained this addition and met the requirements of tables I and II of Federal Specification SS–C–192 had a durability factor of not less than 80 when tested in accordance with the procedures of method 4201.

Note.—Test data showing that an air-entraining addition complies with the requirements of ASTM Designation C–226 will be regarded as acceptable.

3.5 Pozzolan.—Pozzolan which is blended with finished portland cement to produce portland-pozzolan cement shall meet the requirements prescribed in table III. The fineness requirements need not apply to pozzolan which is interground with portland-cement clinker to produce the portland-pozzolan cement. For evaluating the contribution to compressive strength, coarse pozzolans shall be ground only to the extent of having a residue on the No. 325 sieve of 12 ± 2 percent. For materials such as diatomaceous earth with essentially no residue on a No. 325 sieve, the minimum residue need not apply.

TABLE II.—*Physical requirements*

Test	Type I	Type IA
Fineness:		
Residue No. 325 sieve, maximum percent	12	12
Air-permeability:		
Average value, minimum, sq. cm./g	2900	2900
Minimum value, any one sample, in average sq. cm./g	2700	2700
Soundness, autoclave expansion, maximum, percent	0.5	0.5
Time of setting, Gillmore test:		
Initial, minimum, minutes	60	60
Final, maximum, hours	10	10
Compressive strength[1], minimum:		
1 day in moist air, 6 days in water, lb./sq. in.	1500	1250
1 day in moist air, 27 days in water, lb./sq. in.	3000	2500
Air entrainment: percent by volume	0–12	19±3
Water requirement, maximum, ml.	320	280
Drying shrinkage (4.4.5), maximum, percent	0.12	0.11
Mortar expansion:		
At age of 14 days, maximum, percent	0.020	0.020
At age of 8 weeks, max., percent	0.060	0.060
False set[2] early stiffening cement paste, difference between initial and final penetration, not more than mm.	17	17

[1] The strength at 28 days shall be greater than at 7 days.

[2] This requirement applies only when specifically requested in the invitation for bids.

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TABLE III.—*Physical requirements—Pozzolan*

Test	Type P[1]	Type F[2]
Fineness:		
Residue No. 325 sieve, maximum, percent	12.0	12.0
Air-permeability, minimum, sq. cm./g	—	3000
Contribution to compressive strength,[3] (4.5.1):		
Percent of control 28 days, minimum	75	85

[1] Type P refers to natural materials such as clays, shales, diatomaceous earths, tuffs, volcanic ash and pumicite, either calcined or uncalcined. Many natural pozzolans are improved by calcining between 1,400° F. and 1,800° F.

[2] Type F refers to fly ash, an artificial pozzolan which is a fine ash collected from the flue gases at the stacks of power plants burning pulverized coal.

[3] The compressive strength of the portland-pozzolan cubes at 90 days shall not be less than the strength at 28 days.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection. — The manufacturer shall provide suitable facilities to enable the inspector to check the relative weights of portland cement and pozzolan used to produce the portland-pozzolan cement. The plant facilities for blending and inspection shall be adequate to insure compliance with the requirements in 3.2.

4.2 Sampling.—The Government will sample the portland cement and pozzolan prior to blending or the portland-cement clinker and pozzolan prior to intergrinding, and will sample the portland-pozzolan cement. One 5-pound sample of portland-cement or portland-cement clinker shall be taken from approximately each 400 tons. One 5-pound sample shall be taken from approximately each 100 tons of portland-pozzolan cement. Samples of portland cement and samples of portland-pozzolan cement, except samples for false set tests, shall be taken from the conveyor delivering the materials to the bins. A 5-pound sample of portland-pozzolan cement for use in false set tests shall be taken from each car prior to shipment when the purchaser has requested that this test be applied. Samples of portland-cement clinker and of pozzolan shall be taken from bins, or from lots of these materials that have been so stored as to prevent mixing with other lots and prevent contamination.

4.3 The Government will test the portland-cement or portland-cement clinker,[1] the pozzolan and the portland-pozzolan cement. Tests of portland-cement or portland-cement clinker for chemical composition shall be made on composite samples in accordance with Federal Test Method Standard No. 158. Tests of pozzolan for physical properties shall be made on composite samples in accordance with the provision of 4.5. Tests of portland-pozzolan cement for chemical composition and physical properties, except fineness and false set, shall be made on composite samples in accordance with provisions of 4.4. Fineness tests of portland-pozzolan cement shall be made on each 5-pound sample. The test for false set of portland-pozzolan cement (when required) shall be made on each car sample.

4.4 Methods of test, portland-pozzolan cement.—The applicable methods outlined in Federal Test Method Standard No. 158 shall be followed to determine compliance with the provisions of this specification. The following paragraphs furnish detailed test procedures that do not appear in Federal Test Method Standard No. 158, and/or exceptions to these methods.

4.4.1 *Sulfur trioxide.*—The sulfur-trioxide content of the portland-pozzolan cement shall be determined in accordance with method 2 of Federal Test Method Standard No. 158, except that the pozzolan need not be completely decomposed by the acid.

[1] If the portland-cement clinker has been ground in a steel laboratory mill, a magnetic separation of iron particles shall be made prior to chemical analysis.

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4.4.2 *Moisture content.* — Heat a 1-gram sample in a heat-resistant porcelain crucible to constant weight in an oven at 105° C. The percent moisture content equals loss in weight times 100.

4.4.3 *Loss on ignition.*—The loss on ignition shall be determined in accordance with the method given in Federal Test Method Standard No. 158, except that the material remaining from the determination of moisture content shall be ignited to constant weight in the porcelain (not platinum) crucible at a temperature between 700° and 800° C. The weight after heating at 105° C. minus the weight after heating at 700° C. times 100 equals the percent loss in ignition.

4.4.4 *Water requirement.*—The water requirement shall be determined as the milliliters of water required to produce a flow of 110 ± 5 with 25 drops on the flow table.

4.4.5 *Drying shrinkage.*—The mortar for the three shrinkage specimens shall be the same as the mortar used for compressive strength. The molds shall be the same as specified for autoclave test specimens in Federal Test Method Standard No. 158. The specimens shall be molded by placing a small amount of mortar around each reference insert and thoroughly tamping with a 1- by ½- by 6-inch tamper. Fill the molds and tamp 25 times with the above specified tamper held in a vertical position with the 1-inch side of the tamper held parallel to the sides of the mold. This tamping shall be with sufficient force and evenly distributed over the total surface of the bar so that uniform filling of the mold is obtained. After compaction the excess material shall be cut off flush with the top of the mold with a thin-edged trowel and the surface smoothed with a few strokes of the trowel. The shrinkage bars shall be cured 24 hours in the damp closet and then removed from the molds. They shall then be further cured for 6 days in water at 23° C. (73° F.) ± 1.7° C. (3° F). The specimens shall then be read for length and stored at 23° C. (73° F.) ± 1.7° C. (3° F.) and 50 to 55 percent relative humidity. The bars shall again be read for length change at 28 days after their initial reading. The average difference in length of the three specimens between initial and final reading shall be expressed as a percent of the effective gage length and shall be reported to the nearest 0.01 percent.

4.4.6 *Mortar expansion.*—The test for mortar expansion shall be made using molds and comparator as described in Federal Test Method Standard No. 158 for autoclave expansion test.

4.4.6.1 *Storage containers.* — Specimens shall be stored in containers made of noncorrodible material or in containers treated to prevent rusting. Water to a depth of 1 inch shall be provided in the containers at all times and the specimens shall be supported so that they cannot contact the water or the other specimens. Containers shall be sealed. Each container shall have an internal volume no greater than 1 cubic foot.

4.4.6.2 *Pyrex glass.*—Uncontaminated, No. 774, Pyrex glass (obtainable as "lump cullet" from the Corning Glass Works, Corning, N.Y.), crushed, graded, and washed shall be used as aggregate in the fabrication of expansive mortar bars. The grading shall be as follows:

	Percent
Retained on No. 100 sieve	100
Retained on No. 50 sieve	80
Retained on No. 30 sieve	60
Retained on No. 16 sieve	40
Retained on No. 8 sieve	20
Retained on No. 4 sieve	0

4.4.6.3 *Proportioning and mixing.* — The mortar for the mortar-expansion test specimens shall consist of 900 grams of crushed and graded pyrex glass, 400 grams of cement, and enough water to give a flow of 110±5 on the flow table. Mechanical mixing shall not be used. Mixing shall be done in a stainless steel bowl of 5- to 8-quart capacity by vigorous and continuous stirring, squeezing, and kneading with one hand, which shall be protected by a snug-fitting rubber glove. The materials for

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a batch shall be introduced in the following manner:

(1) Place the water in the dry bowl.

(2) Add the cement to the water and mix for 30 seconds.

(3) Add approximately one-half of the graded testing sand and mix for 30 seconds.

(4) Add the remaining sand and mix for 1½ minutes.

4.4.6.4 *Molding of specimens.*—Molds shall be thinly covered with mineral oil; after this operation the reference gage points shall be set in the molds, care being taken to keep them free of oil. Three 1- by 1- by 11¼ inch specimens, having an effective gage length of 10 inches, shall be fabricated from each mortar batch. The specimens shall be molded as described in 4.4.5 for shrinkage bars.

4.4.6.5 *Treatment of specimens.* — After each mold has been filled, it shall be immediately placed in the moist closet and kept there until the specimen has developed sufficient strength for the mold to be removed without injuring the specimen, but, in any event, no longer than 24 hours. After 24 hours curing in the moist closet, the specimens shall be placed in the storage container with a small amount of water. The container shall be sealed and placed in a cabinet at a constant temperature of 38° C. (100° F.) ± 1.7° C. (3° F.) for a period of 13 days. After the 24-hour curing period and before placing in storage, each specimen shall be measured in the comparator for length. When the specimens are 14 days old, they shall be measured again in the comparator for length. The container and specimens shall be cooled at 23° C. (73° F.) ± 1.7° C. (3° F.) for 5 hours before the container is opened and the final reading made. All readings shall be to the nearest 0.0001 inch. The average difference in length of the three specimens between initial and final readings shall be expressed as a percent of the effective gage length and shall be reported to the nearest 0.001 percent. After the readings at 14 days' age have been completed, the specimens shall be returned to the containers and these shall again be sealed and placed in the 38° C. (100° F.) cabinet or room. The specimens shall remain in this storage until they reach an age of 8 weeks when they shall again be measured in the comparator for length. The average difference in their length, between initial and 8 weeks' reading shall again be reported as a percent of the effective gage length.

4.5 **Methods of test, pozzolan.**—Tests for air-permeability fineness and residue retained on the No. 325 sieve shall be made in accordance with the applicable methods outlined in Federal Test Method Standard No. 158.

4.5.1 *Contribution to compressive strength.* —Compressive strengths of 2-inch plastic mortar cubes shall be made for a control mix containing portland cement, and a test mix of the same portland cement and pozzolan for comparison. These cubes shall be fabricated in accordance with Federal Test Method Standard No. 158, except for the following:

4.5.1.1 *Mix parts.*—The mix parts for the control mix shall be one part of any type I cement meeting Federal Specification SS–C–192, 2.75 parts of graded standard Ottawa sand, and sufficient water to obtain a flow of 110 ± 5. The test-mix parts shall be identical to the control-mix parts, except that 35 percent by absolute volume of the control cement shall be replaced with pozzolan, and the water used shall be such as to produce a flow of 110 ± 5. Mortars shall be mixed as directed in Federal Test Method Standard No. 158, Method 2301.

Example:	6 cube batch
Control mix	500 grams cement 1,375 grams Ottawa sand X = ml. of water for flow of 110±5
Test mix	325 grams of cement $175 \times \frac{\text{specific gravity of pozzolan}}{\text{specific gravity of cement}}$ grams of pozzolan 1,375 grams of Ottawa sand Y = ml. of water for flow of 110±5

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4.5.1.2 *Storage of specimens.*—When cube specimens are taken from the molds after 24 hours, they shall be placed in close-fitting containers, sealed airtight, and stored at 38° C. (100° F.) for 27 days, then removed from 38° C. (100° F.) room and stored at 23° C. (73° F.) ± 1.7° C. (3° F.) until the date of test. Cubes broken at 28 days shall be allowed to cool in the containers to 23° C. (73° F.) ± 1.7° C. (3° F.) for 5 hours before being broken. (Any container having a capacity for 3 cubes may be used if it can be made airtight. The use of containers made of light sheet metal with inside dimensions of approximately $2\frac{1}{32}$ by $2\frac{1}{32}$ by $6\frac{1}{4}$ inches is suggested.)

4.5.1.3 *Test ages.*—Compressive strength shall be determined at 28 and 90 days only. Three specimens shall be made for each age. (Ninety-day specimens of the control batch need not be made, but a 6-cube batch shall be mixed.)

4.5.1.4 *Percent of control strength.*—The percent strength is calculated by the following formula:

$$\text{Percent strength} = \frac{\text{average compressive strength of the test mix cubes, in p.s.i.} \times 100}{\text{average compressive strength of the control mix cubes, in p.s.i.}}$$

5. PREPARATION FOR DELIVERY

5.1 For civil agency procurement, the levels of preservation, packaging and packing used are defined in Federal Standard No. 102.

5.2 **Packaging.** — The portland-pozzolan cement shall be delivered in bags, barrels, or in bulk, as specified. A bag shall contain 0.48 cubic foot of material by absolute volume. All packages shall be in good condition at the time of inspection. Packages varying more than 5 percent from the weight corresponding to 0.48 cubic foot may be rejected and if the average weight of packages in any shipment, as shown by weighing 50 packages taken at random, is less than that specified, the entire shipment may be rejected. The portland-pozzolan cement shall be packaged in commercial containers unless otherwise specified by the purchaser in the invitation for bids, contract or order. When shipped in bags, packing shall be in accordance with one of the following levels, as specified.

5.2.1 *Level A.*—Cement, in quantities of 0.48 cubic ft. absolute volume shall be packed in bags conforming to Federal Specification UU–S–48, X series.

5.2.2 *Level B.*—Cement in quantities of 0.48 cubic ft. absolute volume shall be packed in bags conforming to Federal Specification UU–S–48, L/W series.

5.2.3 *Level C.*—Cement shall be packed in a manner which will assure arrival at destination in satisfactory condition and which will be acceptable to the carrier at the lowest transportation cost.

5.3 **Marking.**—When the cement is delivered in packages, the words "Portland-Pozzolan", the name of the manufacturer, and the type of the portland-pozzolan cement, whether it is air-entraining or non-air-entraining, the net weight and absolute volume shall be plainly indicated thereon. Similar information shall be provided in the shipping advices accompanying the shipment of packaged or bulk cement. The manufacturer shall also advise the purchaser of the source of the pozzolan and the amount used in manufacturing the portland-pozzolan cement and the composition of the pozzolan.

5.3.1 *Civil agencies.*—Marking shall be in accordance with Federal Standard 123. When cement is delivered in bags, the name and brand of the manufacturer and the type of cement shall be plainly indicated. Similar information shall be provided in shipping advices accompanying the shipment of packaged or bulk cement.

5.3.2 *Military procurements.* — Marking shall be in accordance with Military Standard MIL–STD–129.

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6. NOTES

6.1 Intended use.—The term "pozzolan" is used to designate siliceous or siliceous and aluminous material which in itself possesses little or no cementitious value but will, in finely divided form and in the presence of moisture, chemically react with calcium hydroxide at ordinary temperatures to form compounds possessing cementitious properties. The term "portland-pozzolan" cement is used to designate an interground mixture of portland-cement clinker, pozzolan and gypsum or an intimate and uniform blend of portland cement and fine pozzolan.

6.2 Ordering data. — Purchasers should specify type required and should exercise any desired options offered herein. (See 1.1, table II and 5.1).

6.3 If purchaser does not specify the type of cement to be furnished, type I will be furnished.

6.4 If the purchaser does not specify that the portland-pozzolan cement be made with either type P or type F pozzolan, the manufacturer may choose for himself the type of pozzolan used. The manufacturer shall state in his proposal what type of pozzolan he will use and be governed by the requirements for that type.

6.5 If the purchaser desires that the portland-pozzolan cement be free of false set, he should specify that the requirements for false set apply in his invitation for bids.

6.6 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.7 Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

6.8 Transportation description. — The proper transportation description applicable is:

Cement, natural or Portland:
Carload weight 50,000 pounds.
Truckload weight 50,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—E
Navy—Y S MC
Air Force

Copies of this specification may be purchased for ~ cents each.

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